NVIDIA GeForce RTX 5070 Graphics Card

Mid-Tier is getting expensive by the generation

With the launch of the RTX 5090, 5080 and 5070 Ti out of the way, NVIDIA has opened the floodgates for the more pragmatic gamers with the launch of the NVIDIA GeForce RTX 5070 graphics card. Why pragmatic? The 70-class graphics cards offer enough performance to comfortably drive high-resolution displays and allow for gamers to experience the best that NVIDIA has to offer without necessarily breaking the bank. This was the experience that the 60-class cards were known for until a few generations ago but has now crept into the 70-class.

7.4

Performance 6.7
Features 7.5
Build Quality 7.8
Value for money... 7.5

All that aside, the 70-class cards are what gamers tend to use for 2-3 generations before even considering an upgrade, which means that GPU manufacturers such as NVIDIA and AMD get quite competitive in this space. This is evident from what we saw with the AMD Radeon RX 9070 XT announcement, wherein AMD has priced the Radeon RX 9070 XT and RX 9070 in such a manner so as to completely eliminate the NVIDIA RTX 5070 Ti and 5070 as viable options for an upgrade. That's of course, only possible if the competing cards can beat the performance of the NVIDIA cards. Things will only become clearer once the reviews are out, so until then, let's have a look at the NVIDIA GeForce RTX 5070 graphics card today.

SPECIFICATIONS

The RTX 5070, built on the NVIDIA Blackwell architecture, features 48 active Streaming Multiprocessors which in turn host 6144 CUDA cores. They're aided by 80 ROPs and 192 Texture Units. The official specifications don't list the ROP and TMU count but the unit we've received has matched up to the claimed performance metrics so we can safely say that our unit was flawless. The base clock of the GPU is 2325 MHz and it can hit a boost clock of 2512 MHz.

As for folks wondering about the ray-tracing and AI capabilities, the RTX 5070 features 48 RT Cores and 192 Tensor Cores which does differentiate the card from the 5070 Ti by a decent margin. So we wouldn't be surprised if NVIDIA launches a couple of Super cards to fill in the performance gaps. Lastly, the card comes with 12 GB of GDDR7 video memory attached to a 192-bit interface reaching a maximum of 28 Gbps or 672 GB/s of effective bandwidth. Again, no surprises if we see an RTX 5070 Super soon.

PERFORMANCE

We have featured some of the older series cards along with most of the RTX 40 series cards including the Super cards. A few older series cards were also thrown into the mix to see how good the cards perform against the competition. In 3DMark Time Spy Extreme, the RTX 5070 is a teensy bit ahead of the RTX 4070 Super and the performance gap between the 5070 and the 5070 Ti is quite proportional to that between the 5070 Ti and the RTX 5080. Even in Ray-Tracing performance, the 5070 takes quite the hit compared to the 5070 Ti. We're obviously talking about pure RTX without the DLSS magic. Even in OpenCL compute benchmarks as well as content creation workloads, the RTX 5070 comes across as a decent graphics card.

GAMING PERFORMANCE

In games such as Assassin's Creed Mirage, The Witcher 3 and F1 24, the RTX 5070 easily provides 60+ FPS at 4K with RTX turned off. So it goes without saying that 1440p and 1080p are handled without breaking a sweat. As for Cyberpunk 2077 and Hogwarts Legacy, the RTX 5070 can't even hit 50 FPS without RTX. That changes immediately when you turn on NVIDIA's greatest gift to mankind. Unfortunately, for those who can easily spot artifacting in video games, the RTX 5070 will give you plenty of opportunities.

POWER CONSUMPTION AND THERMALS

Being a Founders Edition card, the NVIDIA GeForce RTX 5070 that we received has a very tame aesthetic, and